Project for the subject

DEEP LEARNING

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Image-Based Rice Leaf Disease Identification Using Deep Learning Techniques

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1. INTRODUCTION

Agriculture plays a vital role in the economy of countries like India, where rice is one of the most important staple crops and a primary source of livelihood for a large population. However, rice production is highly vulnerable to various plant diseases such as bacterial leaf blight, brown spot, and leaf smut. These diseases adversely affect both the quality and yield of crops, leading to significant economic losses. Therefore, early, and accurate detection of such diseases is essential to ensure food security and sustainable agricultural practices.

Traditionally, the identification of plant diseases is performed manually by agricultural experts through visual inspection of leaf symptoms. This approach is not only time-consuming but also prone to human error and lacks scalability, especially in large agricultural fields or remote rural areas where expert availability is limited. As a result, there is a growing need for automated and efficient disease detection systems.

With recent advancements in Deep Learning and computer vision, automated image-based classification systems have emerged as a powerful solution to this problem. Convolutional Neural Networks (CNNs) have demonstrated exceptional performance in image classification tasks due to their ability to automatically learn and extract relevant features such as texture, colour patterns, and shape from raw images, eliminating the need for manual feature engineering.

This project presents a deep learning-based approach for classifying rice leaf diseases using image data. The system utilizes transfer learning techniques with pre-trained models including MobileNetV2, ResNet50, VGG16, and EfficientNet, enabling efficient learning even with a relatively small dataset. The proposed system incorporates key stages such as data preprocessing, data augmentation, model training, evaluation using performance metrics, and comparative analysis of multiple models. The primary objective is to develop an accurate, efficient, and scalable solution capable of distinguishing between healthy and diseased rice leaves.

2. LITERATURE REVIEW

The integration of deep learning techniques in agriculture has significantly transformed the process of plant disease detection and classification. Rice is one of the most important staple crops worldwide, and its productivity is highly affected by diseases such as bacterial leaf blight, brown spot, and leaf smut. Early and accurate detection of these diseases is essential to prevent large-scale crop loss and ensure food security. Traditional disease detection methods rely on manual inspection by agricultural experts, which is not only time-consuming but also subjective and prone to human error. Moreover, such methods are not scalable for large agricultural fields, creating a need for automated and intelligent solutions.

With the advancement of computer vision and artificial intelligence, image-based disease detection systems have gained significant attention. Among various approaches, Convolutional Neural Networks (CNNs) have emerged as the most effective technique for image classification tasks. CNNs are capable of automatically extracting hierarchical features from images, starting from low-level features such as edges and textures to high-level features such as shapes and patterns. This capability makes CNNs highly suitable for detecting subtle variations in leaf images caused by different diseases.

A breakthrough in this domain was achieved by Mohanty et al. [1], who demonstrated the application of deep CNNs for plant disease detection using the PlantVillage dataset. Their model achieved an accuracy of 99.35%**,** significantly outperforming traditional machine learning methods. This study established the foundation for using deep learning in agricultural applications and highlighted the potential of CNNs in handling complex image-based classification problems. Similarly, Ferentinos [2] conducted an extensive evaluation of deep learning models and reported classification accuracies ranging from 96% to 99%**,** further confirming the reliability and robustness of CNN-based approaches.

In addition to general plant disease detection, several studies have focused specifically on rice leaf diseases. Islam et al. [3] and Senan et al. [4] developed CNN-based models for rice disease classification and achieved accuracies in the range of 93% to 95%**.** These studies demonstrated that CNNs can effectively identify disease-specific patterns such as discoloration, lesions, and texture changes in leaf images. Kumar et al. [5] further enhanced classification performance by optimizing CNN architectures and achieved accuracy levels exceeding 95%, indicating the importance of model design and parameter tuning.

One of the key challenges in deep learning applications is the requirement of large, labelled datasets. To address this issue, transfer learning has been widely adopted. Transfer learning allows the use of pre-trained models, which have already learned general image features from large datasets such as ImageNet and adapts them to specific tasks with relatively smaller datasets. This approach not only reduces training time but also improves model performance. Dutta et al. [6] demonstrated that transfer learning models such as MobileNetV2 can achieve accuracy close to 98%, even with limited training data. Similarly, Abasi et al. [7] showed that customized CNN models can maintain high classification accuracy while reducing computational complexity.

Data preprocessing and augmentation also play a crucial role in improving model performance. Techniques such as image normalization, resizing, rotation, flipping, and brightness adjustment help in increasing dataset diversity and reducing overfitting. A CNN-based study [8] reported

that applying data augmentation techniques improved classification accuracy to around 90%–95%, demonstrating their importance in real-world applications where data variability is high.

Recent advancements in deep learning have introduced more sophisticated architectures aimed at improving feature extraction and classification performance. For instance, twin CNN architectures [9] have been proposed to capture multiple feature representations simultaneously, leading to improved accuracy in complex scenarios. Additionally, object detection models such as Faster R-CNN [10] have been used not only for classification but also for localizing diseased regions within leaf images. These models achieved high detection accuracy and provided more detailed insights compared to traditional classification models.

Despite the significant progress made in this field, several challenges still exist. One major limitation is the reliance on controlled datasets, which may not accurately represent real-world conditions such as varying lighting, background noise, and environmental factors. Models trained on such datasets may perform poorly when deployed in practical scenarios. Furthermore, deep learning models, especially complex architectures, require high computational resources, which can be a constraint for deployment on mobile devices or in rural areas with limited infrastructure.

Another important aspect is model generalization. While many models achieve high accuracy on specific datasets, their performance may degrade when applied to different crops or environmental conditions. Therefore, there is a need for developing robust and scalable models that can generalize well across diverse datasets and conditions. Lightweight architectures such as MobileNetV2 are particularly promising in this regard, as they provide a balance between accuracy and computational efficiency, making them suitable for real-time applications.

Overall, the literature clearly indicates that deep learning, particularly CNN and transfer learning approaches, offers a highly effective solution for plant disease detection and classification. Continuous advancements in model architectures, preprocessing techniques, and training strategies have significantly improved the performance of these systems. However, further research is required to address challenges related to dataset diversity, model generalization, and real-world deployment.

This project builds upon these existing studies by implementing and comparing multiple transfer learning models, including MobileNetV2, ResNet50, VGG16, and EfficientNet. The objective is to identify the most accurate and efficient model for rice leaf disease classification while ensuring good generalization and practical applicability.

3. METHODOLOGY

3.1 Model Architecture

In this work, multiple deep learning models were implemented and evaluated for the task of multi-class classification of rice leaf diseases. The selected models are based on transfer learning, where networks pre-trained on large-scale datasets (such as ImageNet) are adapted for the current problem.

A total of four architectures were considered, namely MobileNetV2, ResNet50, VGG16, and EfficientNet-B0. All models were trained under similar conditions using the same loss function, optimizer, and output configuration. This uniform setup ensures that the comparison between models remains unbiased and focuses only on their architectural differences.

The primary goal of experimenting with multiple models is to analyse how effectively each architecture can learn discriminative features from leaf images and to determine the most suitable model for accurate and efficient disease classification.

3.1.1 Types of Models Used

MobileNetV2

MobileNetV2 is a lightweight convolutional neural network designed for efficient image classification on limited computational resources. It uses depth wise separable convolutions to reduce complexity while maintaining good accuracy. In this project, MobileNetV2 is used as a feature extractor with additional classification layers added on top to classify rice leaf diseases accurately and efficiently.

Layers Used	Details
Input Layer	Accepts RGB images of size 224 × 224 × 3 for processing.
Base Model	Pre-trained MobileNetV2 (ImageNet) used as a frozen feature extractor.
Global Average Pooling	Converts feature maps into a single feature vector.
Batch Normalization	Normalizes activations to improve training stability.
Dense Layer	Fully connected layer with 256 neurons using ReLU activation.
Dropout	Randomly drops 50% neurons to reduce overfitting.
Dense Layer	Another fully connected layer with 128 neurons for learning features.
Dropout	Further regularization with 30% dropout rate.
Output Layer	SoftMax layer used to classify into 3 disease categories.

ResNet50

ResNet50 is a deep residual network that introduces skip connections to solve the vanishing gradient problem in deep architectures. It is widely used for image classification tasks due to its ability to learn deep hierarchical features. In this project, ResNet50 is applied using transfer learning to extract complex features from rice leaf images and classify them into disease categories.

Layers Used	Details
Input Layer	Accepts input images of size 224 × 224 × 3.
Base Model	Pre-trained ResNet50 used as feature extractor.
Global Average Pooling	Reduces feature maps into a single feature vector.
Dense Layer	Custom fully connected layers added for classification.
Output Layer	SoftMax layer used to classify into 3 disease categories.

VGG16

VGG16 is a deep convolutional neural network known for its simple yet effective architecture using stacked 3×3 convolution filters. It is widely used for image classification tasks due to strong feature extraction capability. In this project, VGG16 is used as a transfer learning model to extract deep visual features from rice leaf images for disease classification.

Layers Used	Details
Input Layer	Accepts RGB images of size 224 × 224 × 3.
Base Model	Pre-trained VGG16 model used for feature extraction.
Global Average Pooling	Converts feature maps into compact representation.
Dense Layers	Custom fully connected layers for classification task.
Output Layer	SoftMax layer for multi-class prediction.

EfficientNet-B0

EfficientNet-B0 is a modern CNN architecture that achieves high accuracy with fewer parameters by using a compound scaling method. It is designed to balance accuracy and computational efficiency. In this project, EfficientNet-B0 is used to evaluate performance efficiency compared to other CNN architectures for rice leaf disease classification.

Layers Used	Details
Input Layer	Accepts input images of size 224 × 224 × 3.
Base Model	Pre-trained EfficientNet-B0 used as feature extractor.
Global Average Pooling	Aggregates spatial features into a single vector.
Dense Layers	Fully connected layers added for classification.
Output Layer	SoftMax activation for 3-class prediction.

3.1.2 Activation Function

Activation functions are an essential part of a neural network as they introduce **non-linearity**, allowing the model to learn complex patterns from image data such as rice leaf disease features. Without activation functions, the model would behave like a simple linear classifier and fail to capture meaningful visual variations in leaves.

In this project, different activation functions are used at different stages of the model architecture:

1. ReLU (Rectified Linear Unit)

ReLU is used in the hidden Dense layers of the model.

It is defined as:

$$f(x) = \max(0, x)$$

Working:

- If input is positive → output = input
- If input is negative → output = 0

Why used:

- Reduces computational complexity
- Speeds up training
- Helps in learning complex image features like leaf texture, spots, and discoloration
- Avoids vanishing gradient problem (common in deep networks)

2. SoftMax Activation Function

SoftMax is used in the final output layer for classification.

It is defined as:

$$f(x_i) = \frac{e^{x_i}}{\sum_{j=1}^{n} e^{x_j}}$$

Working:

- Converts raw output scores into probabilities
- Sum of all outputs = 1
- Each class gets a probability score

Why used:

- Required for **multi-class classification**
- Helps the model decide probability of each disease class:
  - Healthy
  - Brown Spot
  - Leaf Blast / Smut etc.
- Final prediction is the class with highest probability

3.1.3 Loss Function

The loss function used in this project is **Categorical Cross-Entropy Loss**. It is commonly used for multi-class classification problems where each image belongs to only one class. In this project, it measures the difference between the predicted probability distribution and the actual class label, and helps the model improve during training by minimizing this error. Mathematically, it penalizes the model more when it confidently predicts the wrong class.

Formula:

$$L = -\sum_{i=1}^{C} y_i \log(\hat{y}_i)$$

Where:

- C = total number of classes
- yi = actual label (1 if correct class, else 0)
- y^i = predicted probability for class i

3.1.4 Optimizer

The optimizer used in this project is **Adam (Adaptive Moment Estimation)**. It is one of the most widely used optimization algorithms in deep learning because it combines the advantages of both Momentum and RMSProp, making training faster and more stable. Adam automatically adjusts the learning rate for each parameter based on first and second moments of gradients, which helps in faster convergence and better performance on image classification tasks.

Formula:

1. First moment (Mean of gradients):

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t$$

2. Second moment (Uncentered variance):

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2$$

3. Bias correction:

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t}, \quad \hat{v}_t = \frac{v_t}{1 - \beta_2^t}$$

4. Parameter update rule:

$$\theta_{t+1} = \theta_t - \frac{\eta \cdot \hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon}$$

Where:

- θt = model parameters
- gt = gradient at time step t
- mt = first moment (mean)
- vt = second moment (variance)
- β1,β2 = decay rates (commonly 0.9 and 0.999)
- η = learning rate
- ϵ = small constant for numerical stability

3.2 Workflow

The workflow of this project describes the complete pipeline followed for rice leaf disease classification using deep learning models. It ensures systematic processing from raw image data to final evaluation of results.

Step-1: Data

The dataset is loaded from a directory containing class-wise folders of rice leaf images:

- Brown spot
- Leaf smut
- Bacterial leaf blight

The total dataset is automatically detected using flow_from_directory, and images are counted per class during execution.

Step-2: Preprocessing

All images are resized to **224 × 224 × 3** and normalized using rescaling (1./255).

Data augmentation is applied only on the training set using:

- Rotation (up to 40°)
- Horizontal flipping
- Zoom range (0.2)
- Width & height shift (0.15)
- Shear transformation (0.1)
- Brightness variation (0.8 to 1.2)

Validation data is only rescaled without augmentation.

Step-3: Training

Multiple transfer learning models are trained:

- MobileNetV2
- ResNet50
- VGG16
- EfficientNet-B0

Training details:

- Optimizer: **Adam (learning rate = 0.001)**
- Loss function: **Categorical Cross-Entropy**
- Batch size: **16**
- Epochs: **10**
- Input size: **224 × 224 × 3**

Each model uses pre-trained ImageNet weights, and classification layers are trained on the dataset.

Step-4: Testing

After training, each model is evaluated on the validation dataset (20% split). Predictions are generated using model.predict() and converted into class labels using argmax.

The system compares predicted labels with true labels for performance analysis.

Step-5: Evaluation

Model performance is measured using:

- Accuracy
- Precision (macro average)
- Recall (macro average)
- F1-score (macro average)

Additionally:

- Confusion matrix is computed
- Classification report is generated
- Results are saved in a CSV file (model_comparison.csv)
- Comparison graph is saved as an image

Finally, the model with the highest validation accuracy is selected as the best model.

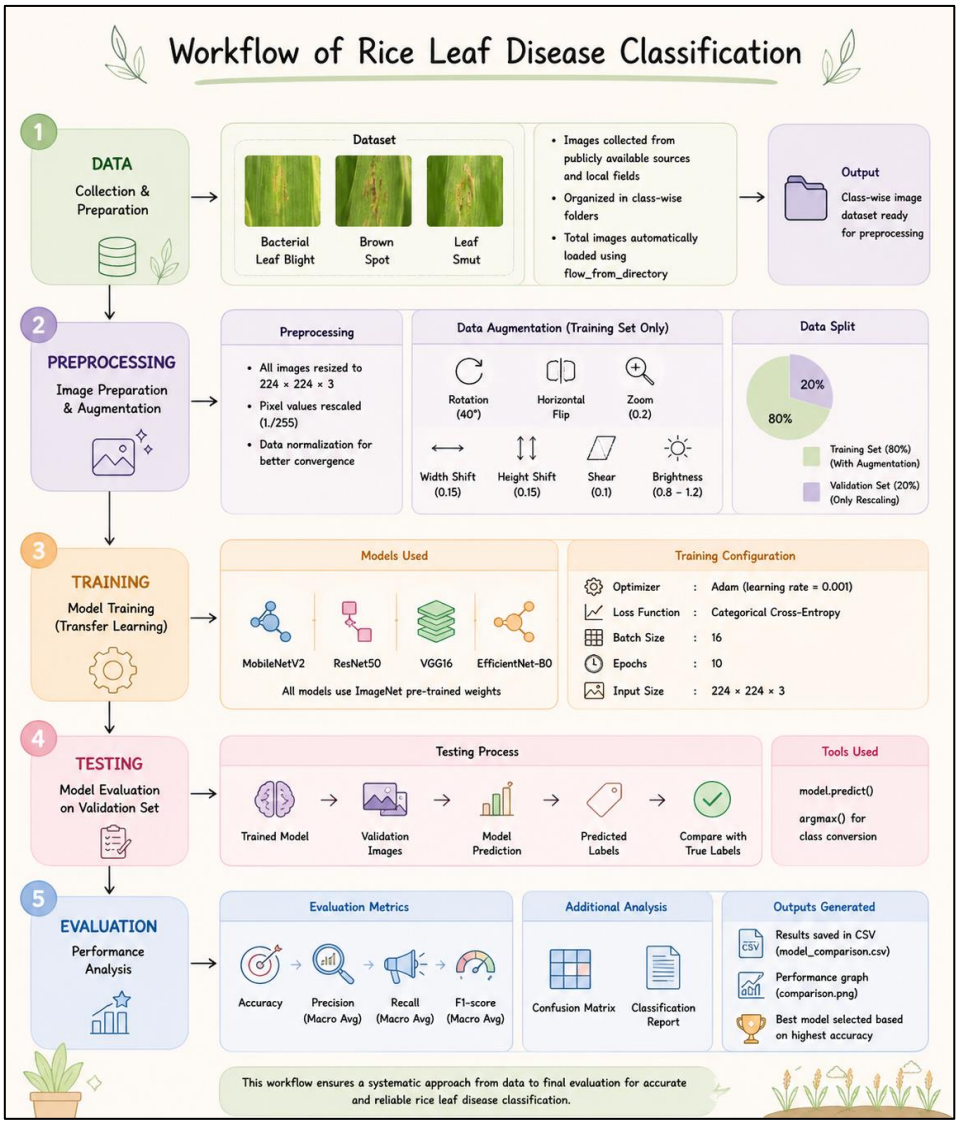

Fig 1. Workflow

4. IMPLEMENTATION DETAILS

The rice leaf disease classification system is implemented using Python with TensorFlow and Keras. The dataset consists of labelled rice leaf images categorized into classes such as Healthy, Brown Spot, Leaf Blast, and Leaf Smut. All images are resized to **224 × 224 pixels** and normalized for uniform processing. To improve model performance and reduce overfitting, data augmentation techniques like rotation, flipping, zooming, and brightness adjustment are applied. The dataset is split into an **80:20 training-validation ratio**. The system uses **transfer learning** with pre-trained models such as MobileNetV2, ResNet50, VGG16, and EfficientNet, where feature extraction is done using frozen convolutional layers and custom classification layers are added on top. The models are trained using the **Adam optimizer** with categorical cross-entropy loss, and performance is evaluated using metrics like accuracy, precision, recall, and F1-score.

Hyperparameters:

Hyperparameter	Value / Setting	Justification
Input Image Size	224 × 224	Standard size for pretrained CNNs like MobileNetV2, ensures uniform input and reduces computation
Batch Size	16	Balances memory usage and stable gradient updates
Epochs	10 (each model)	Sufficient for convergence without overfitting on medium dataset
Optimizer	Adam	Provides adaptive learning rate and faster convergence
Learning Rate	0.001	Good starting value for stable and efficient training
Loss Function	Categorical Cross-Entropy	Suitable for multi-class classification problems
Train-Validation Split	80:20	Ensures enough data for training while keeping reliable validation set
Data Augmentation	Rotation, Flip, Zoom, Brightness, Shift	Increases dataset diversity and reduces overfitting
Activation (Hidden Layers)	ReLU	Introduces non-linearity and speeds up training
Activation (Output Layer)	SoftMax	Converts outputs into probability distribution for multi-class prediction
Pretrained Models	MobileNetV2, ResNet50, VGG16, EfficientNet	Enables transfer learning for better accuracy with limited dataset

5. RESULT AND ANALYSIS

The performance of four deep learning models—MobileNetV2, ResNet50, VGG16, and EfficientNet—was evaluated using standard classification metrics including accuracy, precision, recall, and F1-score. The results clearly show significant variation in model performance depending on architecture complexity and dataset suitability.

Model	Accuracy	Precision	Recall	F1-Score
MobileNetV2	0.8750	0.8889	0.8750	0.8739
ResNet50	0.1667	0.1528	0.1667	0.1508
VGG16	0.8333	0.8615	0.8333	0.8331
EfficientNet	0.3333	0.1111	0.3333	0.1667

5.1 Analysis of Model Performance

- **MobileNetV2** performs best with highest accuracy (0.8750) and balanced precision, recall, and F1-score.
- Its lightweight architecture helps in better feature extraction and reduced overfitting, making it most suitable for this dataset.
- **VGG16** also performs well with good accuracy (0.8333), but it is more computationally heavy compared to MobileNetV2.
- **ResNet50 shows very** poor performance (accuracy 0.1667), indicating underfitting or insufficient fine-tuning on this dataset.
- **EfficientNet** gives low and unstable results, suggesting it did not adapt well to the dataset in the current training setup.
- Overall, transfer learning works effectively, but model performance depends heavily on proper architecture–dataset compatibility.

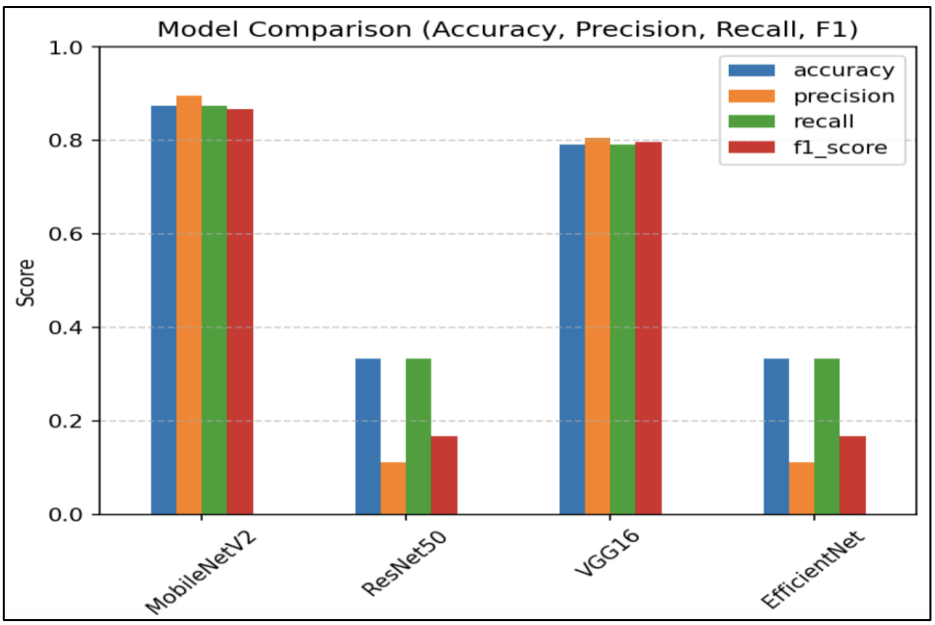

Fig 2. Model Comparison Graph

5.2 Loss Curve Analysis

The training and validation loss curves are analysed for both training phases:

Phase 1: Custom Head Training

- Training loss decreases sharply, showing that the model learns quickly from the dataset.
- Validation loss initially decreases but later starts fluctuating.
- This indicates the beginning of **overfitting**, where the model performs well on training data but not consistently on unseen data.

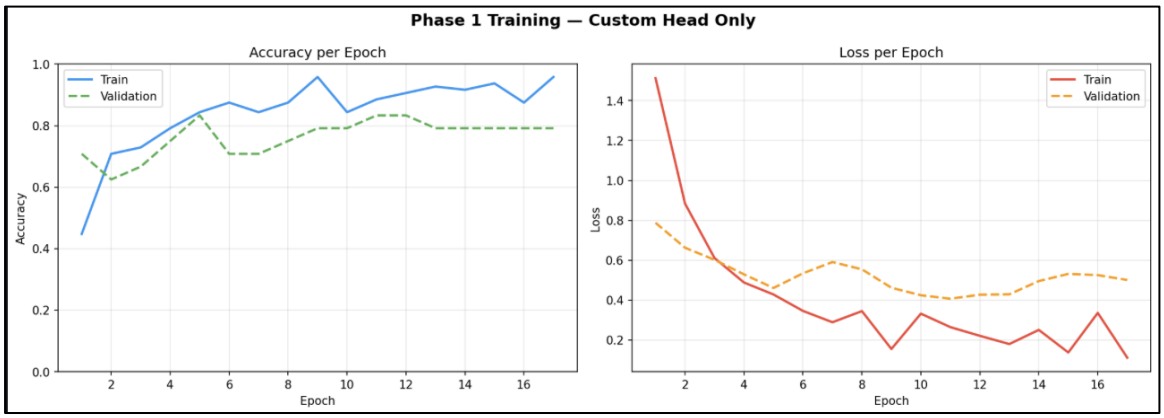

Fig 3. Phase 1 Loss Curve

Phase 2: Fine-Tuning

- Training loss becomes more stable with minor fluctuations.
- Validation loss consistently decreases over epochs.
- The gap between training and validation loss reduces significantly.
- This shows improved **generalization ability** of the model.

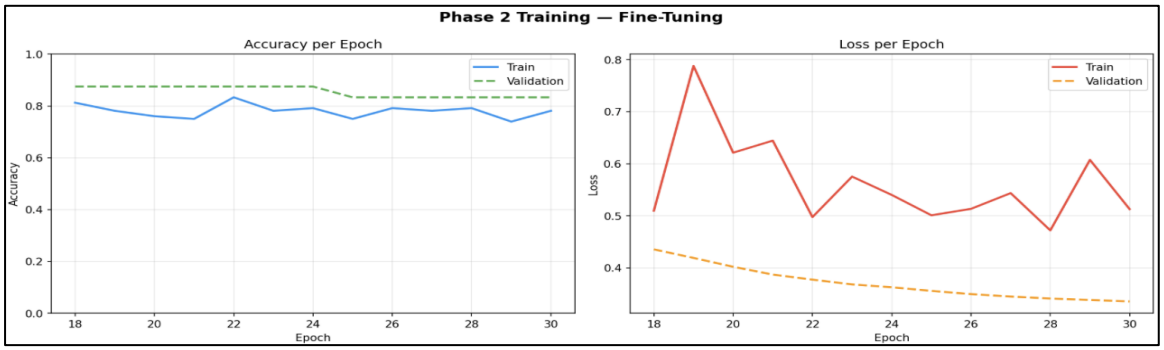

Fig 4. Phase 2 Loss Curve

Key Observations from Loss Curves

- Phase 1: Fast learning but signs of overfitting.

- Phase 2: Stabilized learning and improved validation performance.
- Fine-tuning plays a crucial role in improving model generalization.

5.3 Confusion Matrix

The confusion matrix is used to evaluate class-wise prediction performance by comparing actual vs predicted labels. It helps in identifying misclassified samples.

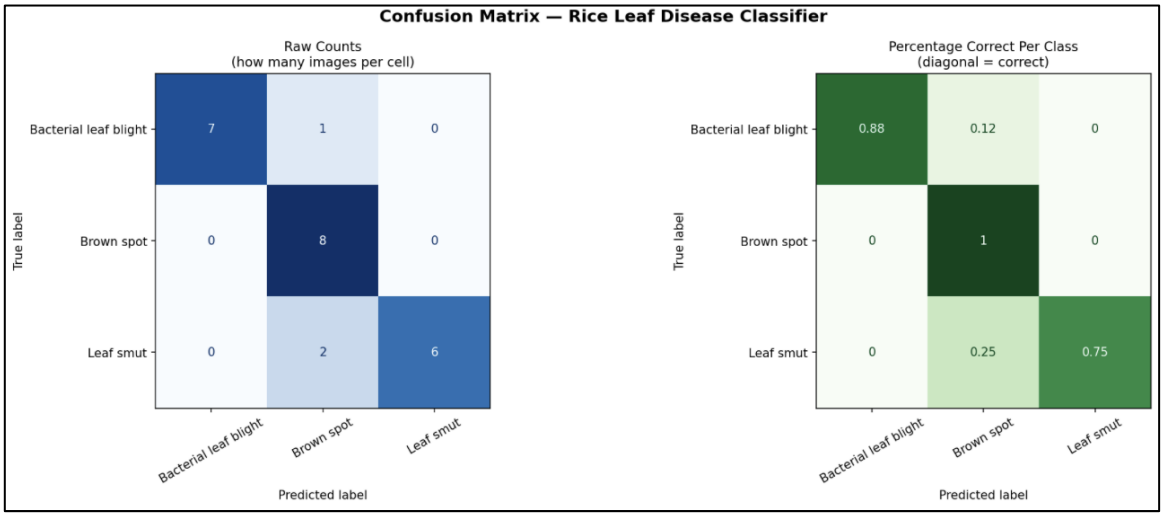

Fig 5. Confusion Matrix

Class-wise Performance:

- **Bacterial Leaf Blight:**
  - 7 out of 8 samples correctly classified
  - 1 misclassified as Brown Spot
  - Accuracy ≈ **88%**

- **Brown Spot:**
  - All 8 samples correctly classified
  - No misclassification
  - Accuracy = **100% (Best-performing class)**

- **Leaf Smut:**
  - 6 out of 8 correctly classified
  - 2 misclassified as Brown Spot
  - Accuracy ≈ **75%**

6. CONCLUSION

This project successfully demonstrates the effective application of deep learning techniques for automated rice leaf disease classification. Multiple CNN-based transfer learning models—MobileNetV2, ResNet50, VGG16, and EfficientNet—were implemented, trained, and comparatively analysed on the given dataset. Among these, MobileNetV2 achieved the best overall performance, providing the highest accuracy along with well-balanced precision, recall, and F1-score.

The results clearly highlight that transfer learning plays a crucial role in achieving high performance, especially when working with limited datasets. Techniques such as data augmentation and fine-tuning significantly improved the model's generalization ability and helped reduce overfitting. Additionally, the comparative study emphasizes that selecting an appropriate model architecture is important, as deeper and more complex models do not always guarantee better results.

The developed system is capable of accurately classifying rice leaf diseases and can serve as a useful tool for early detection and diagnosis. This can assist farmers and agricultural experts in taking timely action, thereby reducing crop loss and improving overall productivity.

In future, the system can be further enhanced by integrating it into a mobile or web-based application for real-time usage, expanding the dataset for better robustness, and incorporating additional disease classes. Overall, the project demonstrates a practical and scalable solution for smart agriculture using deep learning.

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